

Data Base Management System

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Roll No : 15

Batch : B1

Assignment No:5

Table Structure

ProductDetails(ProductID, ProductName, Category, Price, Quantity, SupplierCity)

Perform the following SQL queries:

1. Display all products that belong to the **Electronics category**.

```
mysql> select * from ProductDetails where Category='Electrical';
+-----+-----+-----+-----+-----+-----+
| ProductID | ProductName | Category | Price | Quantity | SupplierCity |
+-----+-----+-----+-----+-----+-----+
| 1003 | TV | Electrical | 3000.00 | 25.00 | Satara |
| 1004 | Fan | Electrical | 1500.00 | 18.00 | Kolhapur |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
```

2. Display products whose **Price is greater than 1000**.

```
mysql> select * from ProductDetails where Price > 1000;
+-----+-----+-----+-----+-----+-----+
| ProductID | ProductName | Category | Price | Quantity | SupplierCity |
+-----+-----+-----+-----+-----+-----+
| 1002 | Pizza Base | FOOD | 2050.00 | 15.00 | Mumbai |
| 1003 | TV | Electrical | 3000.00 | 25.00 | Satara |
| 1004 | Fan | Electrical | 1500.00 | 18.00 | Kolhapur |
| 1005 | Sopa | Furnitures | 6000.00 | 45.00 | Karad |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

3. Display products whose **Quantity is less than 20**.

```
mysql> select * from ProductDetails where Quantity < 20;
+-----+-----+-----+-----+-----+-----+
| ProductID | ProductName | Category | Price | Quantity | SupplierCity |
+-----+-----+-----+-----+-----+-----+
| 1002 | Pizza Base | FOOD | 2050.00 | 15.00 | Mumbai |
| 1004 | Fan | Electrical | 1500.00 | 18.00 | Kolhapur |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

4. Display products that belong to **Furniture or Kitchen** category.

```
mysql> select * from ProductDetails where Category='Furnitures' OR Category='Kitchen';
```

ProductID	ProductName	Category	Price	Quantity	SupplierCity
1005	Sopa	Furnitures	6000.00	45.00	Karad
1006	Cook ware	Kitchen	4000.00	21.00	Thane

2 rows in set (0.00 sec)

5. Display products whose Price is between **500 and 5000**.

```
mysql> select * from ProductDetails where Price BETWEEN 500 AND 5000;
```

ProductID	ProductName	Category	Price	Quantity	SupplierCity
1001	Parle Biscuits	FOOD	1000.00	35.00	Pune
1002	Pizza Base	FOOD	2050.00	15.00	Mumbai
1003	TV	Electrical	3000.00	25.00	Satara
1004	Fan	Electrical	1500.00	18.00	Kolhapur
1006	Cook ware	Kitchen	4000.00	21.00	Thane

5 rows in set (0.00 sec)

6. Display all product records in **ascending order of ProductName**.

```
mysql> select * from ProductDetails ORDER BY ProductName;
```

ProductID	ProductName	Category	Price	Quantity	SupplierCity
1006	Cook ware	Kitchen	4000.00	21.00	Thane
1004	Fan	Electrical	1500.00	18.00	Kolhapur
1001	Parle Biscuits	FOOD	1000.00	35.00	Pune
1002	Pizza Base	FOOD	2050.00	15.00	Mumbai
1005	Sopa	Furnitures	6000.00	45.00	Karad
1003	TV	Electrical	3000.00	25.00	Satara

6 rows in set (0.00 sec)

7. Display all product records in **descending order of Price**.

```
mysql> select * from ProductDetails ORDER BY ProductName DESC;
```

ProductID	ProductName	Category	Price	Quantity	SupplierCity
1003	TV	Electrical	3000.00	25.00	Satara
1005	Sopa	Furnitures	6000.00	45.00	Karad
1002	Pizza Base	FOOD	2050.00	15.00	Mumbai
1001	Parle Biscuits	FOOD	1000.00	35.00	Pune
1004	Fan	Electrical	1500.00	18.00	Kolhapur
1006	Cook ware	Kitchen	4000.00	21.00	Thane

6 rows in set (0.00 sec)

8. Find the **total number of products** in the table.

```
mysql> select Count(productName) as Total_Products from ProductDetails;
+-----+
| Total_Products |
+-----+
|          6 |
+-----+
1 row in set (0.00 sec)
```

9. Find the **total quantity of all products**.

```
mysql> select SUM(Quantity) as Total_Quantity from ProductDetails;
+-----+
| Total_Quantity |
+-----+
|         159.00 |
+-----+
1 row in set (0.00 sec)
```

10. Find the **average price** of all products.

```
mysql> select AVG(Price) as Average_Price from ProductDetails;
+-----+
| Average_Price |
+-----+
|  2925.000000 |
+-----+
1 row in set (0.00 sec)
```

11. Find the **highest price** among all products.

```
mysql> select MAX(Price) as Max_Price from ProductDetails;
+-----+
| Max_Price |
+-----+
|   6000.00 |
+-----+
1 row in set (0.03 sec)
```

12. Find the **lowest price** among all products.

```
mysql> select MIN(Price) as Low_Price from ProductDetails;
+-----+
| Low_Price |
+-----+
|   1000.00 |
+-----+
1 row in set (0.00 sec)
```

13. Find the **number of products in each category**.

```
mysql> select * from ProductDetails;
+-----+-----+-----+-----+-----+-----+
| ProductID | ProductName | Category | Price | Quantity | SupplierCity |
+-----+-----+-----+-----+-----+-----+
| 1001 | Parle Biscuits | FOOD | 1000.00 | 35.00 | Pune |
| 1002 | Pizza Base | FOOD | 2050.00 | 15.00 | Mumbai |
| 1003 | TV | Electrical | 3000.00 | 25.00 | Satara |
| 1004 | Fan | Electrical | 1500.00 | 18.00 | Kolhapur |
| 1005 | Sopa | Furnitures | 6000.00 | 45.00 | Karad |
| 1006 | Cook ware | Kitchen | 4000.00 | 21.00 | Thane |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> select Category,Count(ProductName) as Total_Products from ProductDetails GROUP BY Category;
+-----+-----+
| Category | Total_Products |
+-----+-----+
| FOOD | 2 |
| Electrical | 2 |
| Furnitures | 1 |
| Kitchen | 1 |
+-----+-----+
4 rows in set (0.00 sec)
```

14.Find the **average price of products in each category.**

```
mysql> select Category,AVG(Price) as Average from ProductDetails GROUP BY Category;
+-----+-----+
| Category | Average |
+-----+-----+
| FOOD | 1525.000000 |
| Electrical | 2250.000000 |
| Furnitures | 6000.000000 |
| Kitchen | 4000.000000 |
+-----+-----+
4 rows in set (0.00 sec)
```

15.Find the **total quantity of products in each category.**

```
mysql> select Category,SUM(Quantity) as Total_Quantity from ProductDetails GROUP BY Category;
+-----+-----+
| Category | Total_Quantity |
+-----+-----+
| FOOD | 50.00 |
| Electrical | 43.00 |
| Furnitures | 45.00 |
| Kitchen | 21.00 |
+-----+-----+
4 rows in set (0.00 sec)
```

16.Display only those categories having **more than 2 products**.

```
mysql> select * from ProductDetails;
+-----+-----+-----+-----+-----+-----+
| ProductID | ProductName | Category | Price | Quantity | SupplierCity |
+-----+-----+-----+-----+-----+-----+
| 1001 | Parle Biscuits | FOOD | 1000.00 | 35.00 | Pune |
| 1002 | Pizza Base | FOOD | 2050.00 | 15.00 | Mumbai |
| 1003 | TV | Electrical | 3000.00 | 25.00 | Satara |
| 1004 | Fan | Electrical | 1500.00 | 18.00 | Kolhapur |
| 1005 | Sopa | Furnitures | 6000.00 | 45.00 | Karad |
| 1006 | Cook ware | Kitchen | 4000.00 | 21.00 | Thane |
| 1007 | Sweats | FOOD | 1300.00 | 18.00 | Radhanagari |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)
```

```
mysql> select Category,Count(ProductName) as Total_Products from ProductDetails GROUP BY Category Having Count(*) > 2;
+-----+-----+
| Category | Total_Products |
+-----+-----+
| FOOD | 3 |
+-----+-----+
1 row in set (0.00 sec)
```

17.Display categories where the **average price is greater than 2000**.

```
mysql> select Category,AVG(Price) as Average_Price from ProductDetails GROUP BY Category Having AVG(Price) > 2000;
+-----+-----+
| Category | Average_Price |
+-----+-----+
| Electrical | 2250.000000 |
| Furnitures | 6000.000000 |
| Kitchen | 4000.000000 |
+-----+-----+
3 rows in set (0.00 sec)
```

18.Display all the **different SupplierCity** values available in the ProductDetails table.

```
mysql> Select DISTINCT SupplierCity from ProductDetails;
+-----+
| SupplierCity |
+-----+
| Pune |
| Mumbai |
| Satara |
| Kolhapur |
| Karad |
| Thane |
| Radhanagari |
+-----+
7 rows in set (0.01 sec)
```

19. Find the **number of products supplied from Pune**.

```
ERROR 1054 (42S22): Unknown column 'City' in 'where clause'
mysql> select Count(ProductName) as Total_Products from ProductDetails where SupplierCity = 'Pune';
+-----+
| Total_Products |
+-----+
|                2 |
+-----+
1 row in set (0.00 sec)

mysql>
```

20. Display the **category-wise product count and average price**, showing only categories having more than 1 product, arranged by product count in descending order.

```
mysql> Select Category,Count(*) as Total_Products,AVG(Price) from ProductDetails GROUP BY Category Having Count(*) > 1 ORDER BY Total_Products DESC;
+-----+-----+-----+
| Category | Total_Products | AVG(Price) |
+-----+-----+-----+
| FOOD     | 3              | 1450.000000 |
| Electrical | 2              | 2250.000000 |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

Table Structure

BankTransaction(TransactionID, CustomerName, AccountType, TransactionType, Amount, BranchCity)

Perform the following SQL queries:

1. Display all transactions made by customers having **Savings** accounts.

```
MariaDB [demo]> select * from BankTransaction;
```

TransactionID	CustomerName	AccountType	TransactionType	Amount	City
10011	Harshad Teli	Saving	Deposit	56000	Radhanagari
10012	Shrinath Teli	Current	Deposit	4000	Kolhapur
10013	Anjali Patil	Saving	Withdrwal	85000	Pune
10014	Raj More	FD	Withdrawal	75000	Mumbai
10015	Nandini Sutar	Saving	Deposit	120000	Pune
10016	Samarth Patil	FD	Withdrawal	48000	Karad

```
6 rows in set (0.001 sec)
```

```
MariaDB [demo]> select * from BankTransaction where AccountType='Saving';
```

TransactionID	CustomerName	AccountType	TransactionType	Amount	City
10011	Harshad Teli	Saving	Deposit	56000	Radhanagari
10013	Anjali Patil	Saving	Withdrwal	85000	Pune
10015	Nandini Sutar	Saving	Deposit	120000	Pune

```
3 rows in set (0.002 sec)
```

2. Display transactions where the **Amount is greater than 50,000**.

```
MariaDB [demo]> select * from BankTransaction where Amount > 50000;
```

TransactionID	CustomerName	AccountType	TransactionType	Amount	City
10011	Harshad Teli	Saving	Deposit	56000	Radhanagari
10013	Anjali Patil	Saving	Withdrwal	85000	Pune
10014	Raj More	FD	Withdrawal	75000	Mumbai
10015	Nandini Sutar	Saving	Deposit	120000	Pune

```
4 rows in set (0.001 sec)
```

3. Display transactions where the **Amount is less than 10,000**.

```
MariaDB [demo]> select * from BankTransaction where Amount < 10000;
```

TransactionID	CustomerName	AccountType	TransactionType	Amount	City
10012	Shrinath Teli	Current	Deposit	4000	Kolhapur

```
1 row in set (0.001 sec)
```

4. Display transactions having **Deposit or Withdrawal** as the TransactionType.

```
MariaDB [demo]> select * from BankTransaction where TransactionType='Deposit' OR 'Withdrawal';
```

TransactionID	CustomerName	AccountType	TransactionType	Amount	City
10011	Harshad Teli	Saving	Deposit	56000	Radhanagari
10012	Shrinath Teli	Current	Deposit	4000	Kolhapur
10015	Nandini Sutar	Saving	Deposit	120000	Pune

```
3 rows in set, 5 warnings (0.001 sec)
```

5. Display transactions where the Amount is between **10,000 and 75,000**.

```
MariaDB [demo]> select * from BankTransaction where Amount BETWEEN 10000 AND 75000;
```

TransactionID	CustomerName	AccountType	TransactionType	Amount	City
10011	Harshad Teli	Saving	Deposit	56000	Radhanagari
10014	Raj More	FD	Withdrawal	75000	Mumbai
10016	Samarth Patil	FD	Withdrawal	48000	Karad

```
3 rows in set (0.001 sec)
```

6. Display all transaction records in **ascending order of CustomerName**.

```
MariaDB [demo]> select * from BankTransaction ORDER BY CustomerName ASC;
```

TransactionID	CustomerName	AccountType	TransactionType	Amount	City
10013	Anjali Patil	Saving	Withdrwal	85000	Pune
10011	Harshad Teli	Saving	Deposit	56000	Radhanagari
10015	Nandini Sutar	Saving	Deposit	120000	Pune
10014	Raj More	FD	Withdrawal	75000	Mumbai
10016	Samarth Patil	FD	Withdrawal	48000	Karad
10012	Shrinath Teli	Current	Deposit	4000	Kolhapur

```
6 rows in set (0.001 sec)
```

7. Display all transaction records in **descending order of Amount**.

```
MariaDB [demo]> select * from BankTransaction ORDER BY Amount DESC;
```

TransactionID	CustomerName	AccountType	TransactionType	Amount	City
10015	Nandini Sutar	Saving	Deposit	120000	Pune
10013	Anjali Patil	Saving	Withdrwal	85000	Pune
10014	Raj More	FD	Withdrawal	75000	Mumbai
10011	Harshad Teli	Saving	Deposit	56000	Radhanagari
10016	Samarth Patil	FD	Withdrawal	48000	Karad
10012	Shrinath Teli	Current	Deposit	4000	Kolhapur

```
6 rows in set (0.001 sec)
```

8. Find the **total number of transactions** in the table.

```
MariaDB [demo]> select count(TransactionID) as Total_Transactions from Banktransaction;
+-----+
| Total_Transactions |
+-----+
|          6 |
+-----+
1 row in set (0.001 sec)
```

9. Find the **total transaction amount**.

```
MariaDB [demo]> select SUM(Amount) as Total_Amount from Banktransaction;
+-----+
| Total_Amount |
+-----+
|      388000 |
+-----+
1 row in set (0.001 sec)
```

10. Find the **average transaction amount**.

```
MariaDB [demo]> select AVG(Amount) as Average_Amount from Banktransaction;
+-----+
| Average_Amount |
+-----+
|    64666.6667 |
+-----+
1 row in set (0.001 sec)
```

11. Find the **highest transaction amount**.

```
MariaDB [demo]> select MAX(Amount) as Highest_Amount from BankTransaction;
+-----+
| Highest_Amount |
+-----+
|      120000 |
+-----+
1 row in set (0.001 sec)
```

12. Find the **lowest transaction amount**.

```
MariaDB [demo]> select MIN(Amount) as Lowest_Amount from BankTransaction;
+-----+
| Lowest_Amount |
+-----+
|         4000 |
+-----+
1 row in set (0.001 sec)
```

13. Find the **number of transactions for each AccountType**.

```
MariaDB [demo]> select AccountType,count(TransactionID) as Total_Transactions from BankTransaction GROUP BY AccountType;
```

AccountType	Total_Transactions
Current	1
FD	2
Saving	3

```
3 rows in set (0.001 sec)
```

14. Find the **average transaction amount** for each AccountType.

```
MariaDB [demo]> select AccountType,AVG(Amount) as Average_Amount from BankTransaction GROUP BY AccountType;
```

AccountType	Average_Amount
Current	4000.0000
FD	61500.0000
Saving	87000.0000

```
3 rows in set (0.002 sec)
```

15. Find the **total transaction amount** for each TransactionType.

```
MariaDB [demo]> select AccountType,SUM(Amount) as Total_Amount from BankTransaction GROUP BY AccountType;
```

AccountType	Total_Amount
Current	4000
FD	123000
Saving	261000

```
3 rows in set (0.002 sec)
```

16. Display only those AccountTypes having **more than 2 transactions**.

```
MariaDB [demo]> select AccountType,Count(TransactionID) as Total_Transactions from BankTransaction GROUP BY AccountType HAVING Count(*) > 2;
```

AccountType	Total_Transactions
Saving	3

```
1 row in set (0.002 sec)
```

17. Display TransactionTypes where the **average transaction amount is greater than 25,000**.

```
MariaDB [demo]> select AccountType,AVG(Amount) as Average_Amount from BankTransaction GROUP BY AccountType HAVING AVG(Amount) > 25000;
```

AccountType	Average_Amount
FD	61500.0000
Saving	87000.0000

```
2 rows in set (0.002 sec)
```

18. Display all the **different BranchCity** values available in the table.

```
MariaDB [demo]> select DISTINCT City from BankTransaction;
```

City
Radhanagari
Kolhapur
Pune
Mumbai
Karad

```
5 rows in set (0.001 sec)
```

19. Find the **number of transactions made from Pune branch.**

```
MariaDB [demo]> select count(TransactionID) as Total_Transactions from BankTransaction where City='Pune';
```

Total_Transactions
2

```
1 row in set (0.001 sec)
```

20. Display the **AccountType-wise transaction count and average transaction amount**, showing only account types having more than 1 transaction, arranged by transaction count in descending order.

```
MariaDB [demo]> select AccountType,Count(TransactionID) as Total_Transactions ,AVG(Amount) as Average_Amount from BankTransaction GROUP BY AccountType Having Count(*) > 1 ORDER BY Total_Transactions DESC;
```

AccountType	Total_Transactions	Average_Amount
Saving	3	87000.0000
FD	2	61500.0000

```
2 rows in set (0.002 sec)
```

